

Robotic Arm Documentation

A robotic arm is a type of mechanical arm, usually programmable, with similar functions to a human arm; the arm may be the sum total of the mechanism or may be part of a more complex robot. The links of such a manipulator are connected by joints allowing either rotational motion (such as in an articulated robot) or translational (linear) displacement. The links of the manipulator can be considered to form a kinematic chain. The terminus of the kinematic chain of the manipulator is called the **end effector** and it is **analogous to the human hand**. [WIKIPEDIA]

Components of a Robotic Arm

CONTROLLER

Proportional Integral Derivative Controller (PID Controller)

The Proportional Integral Derivative Controller, commonly known as the PID Controller, is the most widely used type of controller in industrial control systems. It combines the features of both P and PI Controllers and adds a derivative term.

This additional term predicts future errors based on the rate of change of the error, allowing the control system to react to changes more quickly and effectively. PID Controllers provide excellent control performance in many different applications, but they require careful tuning to work optimally.

Fuzzy Logic Controllers

Fuzzy Logic Controllers operate on a different principle altogether. They use fuzzy logic, a form of multi-value logic that allows for reasoning in an uncertain or fuzzy environment. Fuzzy Logic Controllers can handle non-linear systems and systems where the mathematical model is not known or too complex to derive.

They are capable of making decisions based on linguistic information and are often used in applications like automatic washing machines, for example where the input parameters can vary significantly. Many modern robotic controllers are equipped with powerful 32-bit processors.

Neural Network Controllers

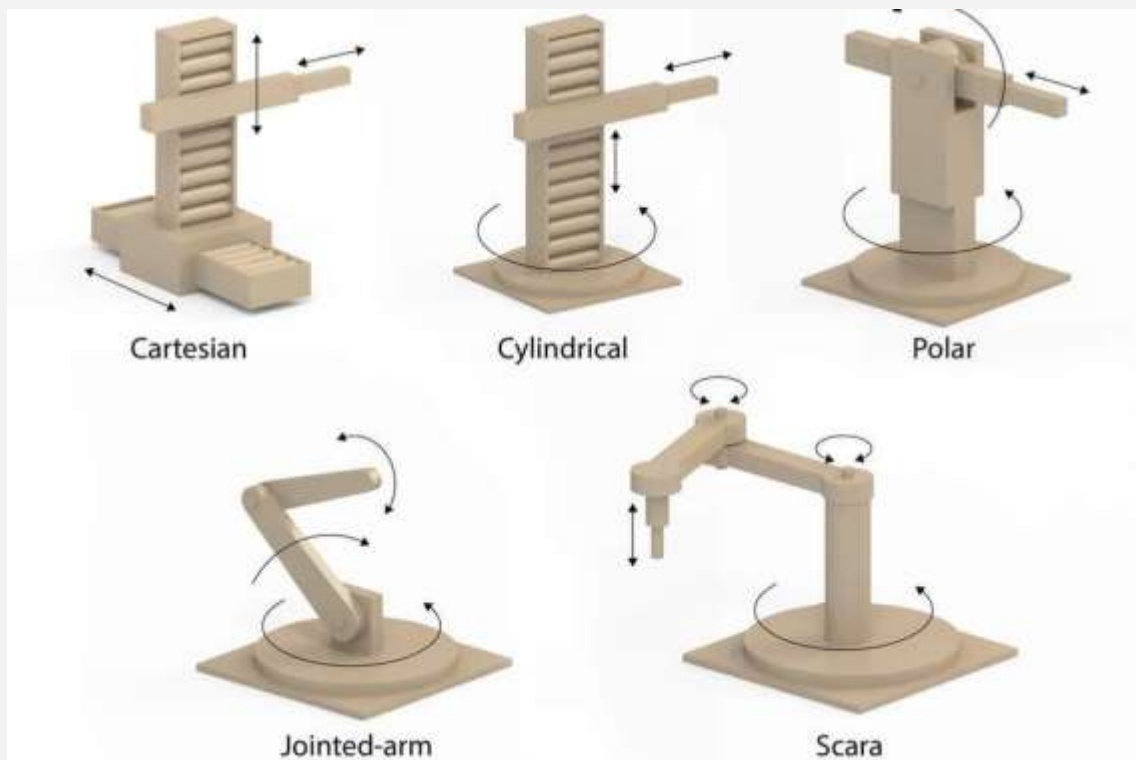
Neural Network Controllers are inspired by the human brain's processing capabilities. These controllers learn from experience, adapting their control strategy based on the data they process. Neural Network Controllers can handle highly complex, non-linear systems and can learn to control systems where other types of controllers might struggle. However, they require large amounts of data and computational resources to function effectively.

ARTICULATIONS AND MID JOINTS

- Common joint types: revolute (rotational), prismatic (linear), and spherical.

END EFFECTOR

- Examples: grippers, welding torches, cameras, or laser cutters.



Top Three Robot Programming Methods [<https://blog.robotiq.com>]

1. Teaching Pendant

The robot teaching pendant has changed a lot throughout its lifetime, but often consists of, what looks like, a giant handheld calculator. Early pendants were large, grey boxes with magnetic tape storage. The modern teach pendants are more like a touchscreen tablet, as the technology has developed to suit the ever-evolving users. To program the robot, the operator moves it from point-to-point, using the buttons on the pendant to move it around and save each position individually. When the whole program has been learned, the robot can play back the points at full speed.

2. Simulation/Offline Programming (RoboDK)

3. Teaching by Demonstration

Teaching by demonstration (and more specific methods like Kinetiq teaching) offers an intuitive addition to the classic teach pendant. These methods involve moving the robot around, either by manipulation a force sensor or a joystick attached to the robot wrist just above the end effector.

Applications [<https://www.universal-robots.com/in/blog/robotic-arm/>]

1. **Manufacturing and Assembly:**

- PCB components assembly.
- Automotive assembly lines.
- Precision machining.

2. **Medical:**

- Surgical procedures (laparoscopy, orthopedics).
- Drug dispensing.

3. **Electronics:**

- Soldering.
- PCB inspection and repair.

4. **Aerospace:**

- Aircraft assembly.
- Satellite deployment.

5. **Research and Development:**

- Prototyping.
- Testing new designs.